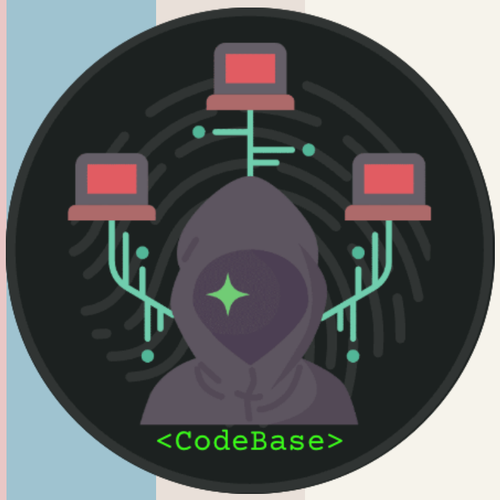
SMART INDIA HACKATHON



- **Problem Statement ID :** SIH25150
- **Problem Statement Title :** Beneficiary Credit Scoring with Income Verification Layer for Direct Digital Lending
- **Team Name :** Code-Base
- **Team ID :** 73668

VISUAL REPRESENTATION OF PROBLEM

Scattered Proofs



Manual Checks



Informal Income



Data Gaps







Delays & Drop-offs



Unfair Rejections

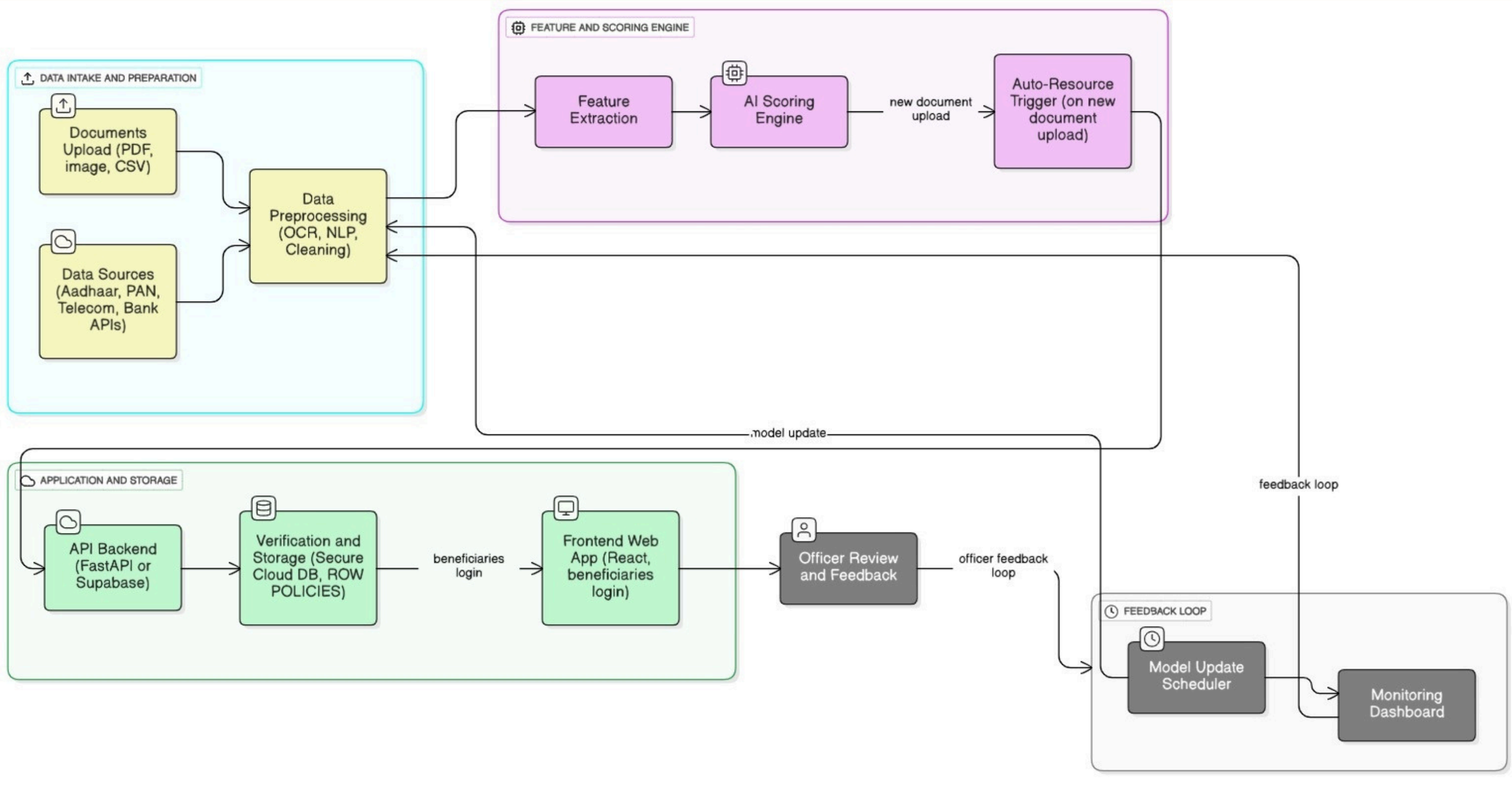


Low Trust



Leakage & Cost

OVERALL WORKFLOW OF THE SOLUTION



MODULES OVERVIEW

1. Data Ingestion Module
2. Preprocessing and Income Verification Module
3. Credit Scoring & Risk Band Module
4. Lending & Dashboard Module

Data Ingestion Module

- Collects user-submitted docs (Aadhaar, PAN, bank statements).
- API links to NBCFDC & UIDAI for real-time fetch.
- Ensures data integrity and standard format conversion.

Preprocessing & Income Verification Module

- Cleans and normalizes inputs using Pandas.
- Extracts income patterns from bank transactions.
- Flags missing or inconsistent records for review.

Credit Scoring & Risk Band Module

- ML model assigns Composite Credit Score (0–900).
- Combines repayment (40%) + income (30%) + savings (20%) + digital usage (10%).
- Categorizes into Low/Medium/High risk bands.

Lending & Dashboard Module

- Auto-approves low-risk applications for same-day disbursal.
- Officer dashboard for verification & analytics.
- Tracks beneficiary performance & feedback loop.

WHAT MAKES SMARTCREDITAI DIFFERENT?

Aspects	Traditional NBCFDC Process	SmartCreditAI
Assessment Time	2–4 weeks (manual verification)	Same-day approval (auto-verification)
Credit Scoring	No standardized method	AI-based explainable score (0–900)
Data Sources	Physical documents only	Multi-modal (bank, Aadhaar, utility data)
Transparency	Subjective officer decisions	Explainable AI with visible weightage
Scalability	Limited by manpower	Cloud-native, multi-tenant scalability
Fraud Detection	Manual spotting (slow)	Real-time anomaly detection
Accessibility	Office visits mandatory	Mobile + Web, multilingual UI
Continuous Improvement	Static yearly audits	Quarterly ML retraining

IMPACT ANALYSIS



BENEFICIARIES

Same-day loan access with transparent, explainable scores.



BANKS/MFIs

Automated risk assessment reduces default risk & manual effort.



GOVERNMENT

Enables policy planning via centralized, data-driven dashboards.



MACRO IMPACT

Promotes large-scale financial inclusion and credit awareness.

Long Term Growth Plan

Mobile App for Beneficiaries



Direct, self-service access for applying & checking eligibility

Multi-Scheme Integration



Auto-fetch data from JanSamarth, PMEGP & other welfare portals.

Voice-Based Interface



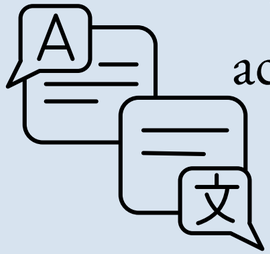
Local-language onboarding for rural / low-literacy users.

Blockchain Ledger



Ensures audit-ready, tamper-proof transaction history.

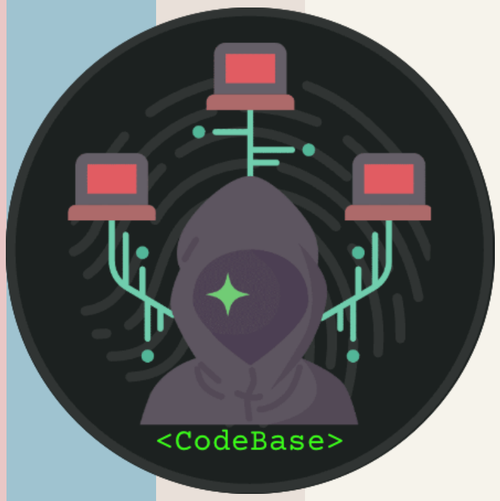
Multilingual Dashboard



Expands reach across 10+ Indian languages.

GITHUB REPOSITARY :

https://github.com/Sree-2607/smart-credit-ai



THANK YOU

SmartCreditAI — Empowering Financial Inclusion through Explainable AI.